

Here are the most likely questions your examiner will ask, with strong answers:

Q1: What is the objective of your project? To detect crop diseases in Arecanut and Tomato plants using AI by analyzing leaf images, and provide farmers with treatment and prevention recommendations instantly.

Q2: Which algorithm/model did you use? We used MobileNetV2, a Convolutional Neural Network (CNN) architecture, with Transfer Learning. It was pre-trained on ImageNet and fine-tuned on our crop disease dataset.

Q3: What is Transfer Learning? Transfer Learning means reusing a model already trained on a large dataset (ImageNet with 1.2 million images) and adapting it to our specific task. This saves training time and improves accuracy even with a smaller dataset.

Q4: What is the accuracy of your model?

- Validation Accuracy: **90.28%**
 - Test Accuracy: **89.00%**
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Q5: How many classes does your model detect? 9 classes — 4 Arecanut diseases (Healthy, BudRot, Koleroga, YellowLeaf) and 5 Tomato diseases (Healthy, EarlyBlight, LateBlight, LeafCurl, SeptoriaLeafSpot).

Q6: What technologies did you use?

- **AI/ML:** PyTorch, MobileNetV2
 - **Backend:** FastAPI, Python 3.12
 - **Frontend:** React.js
 - **Database:** MySQL (XAMPP), SQLAlchemy ORM
 - **Image Processing:** PIL (Pillow), torchvision
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Q7: Why did you choose MobileNetV2? MobileNetV2 is lightweight and fast, making it suitable for devices with limited resources like mobile phones and laptops. It gives good accuracy with less computation compared to heavier models like VGG or ResNet.

Q8: What is the difference between training, validation, and test sets?

- **Train (70%):** Used to teach the model
 - **Validation (20%):** Used to tune the model and prevent overfitting
 - **Test (10%):** Used to evaluate final real-world performance on unseen data
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Q9: What is overfitting and how did you prevent it? Overfitting is when the model memorizes training data but fails on new data. We prevented it using:

- Data Augmentation (rotation, flipping, color jitter)
 - Dropout layers in the classifier
 - Early stopping via ReduceLROnPlateau scheduler
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Q10: What is the role of FastAPI? FastAPI acts as the backend server. It receives the leaf image from the frontend, passes it to the AI model for prediction, and returns the disease name, confidence score, and treatment information as a JSON response.

Q11: What is the role of the database? MySQL stores user information, crop and disease details, scan history, and farmer feedback. SQLAlchemy ORM maps Python classes to database tables.

Q12: What is CORS and why did you use it? CORS (Cross-Origin Resource Sharing) allows the React frontend running on port 3000 to communicate with the FastAPI backend on port 8001. Without it, the browser would block the API calls for security reasons.

Q13: What are the limitations of your project?

- Works only for Arecanut and Tomato crops currently
- Requires good quality leaf images for accurate detection

- Arecanut Koleroga had very few training images (only 10) which may affect accuracy
 - Needs internet to download the pre-trained model the first time
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Q14: What are the future enhancements?

- Add more crops like Coconut, Paddy, Sugarcane
 - Deploy as a mobile app for offline use by farmers
 - Add multilingual support (Kannada, Tamil, Hindi)
 - Integrate weather data for disease risk prediction
 - Retrain model with more images to improve accuracy
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Q15: What is Softmax and how is it used here? Softmax converts the model's raw output (logits) into probabilities that sum to 100%. The class with the highest probability is the predicted disease. For example, 83.91% for SeptoriaLeafSpot.

Memorize these well and you'll handle any question confidently! 💪 🌱 All the best! 😊